Monocular 3D reconstruction

September 2024

Alberto Rota

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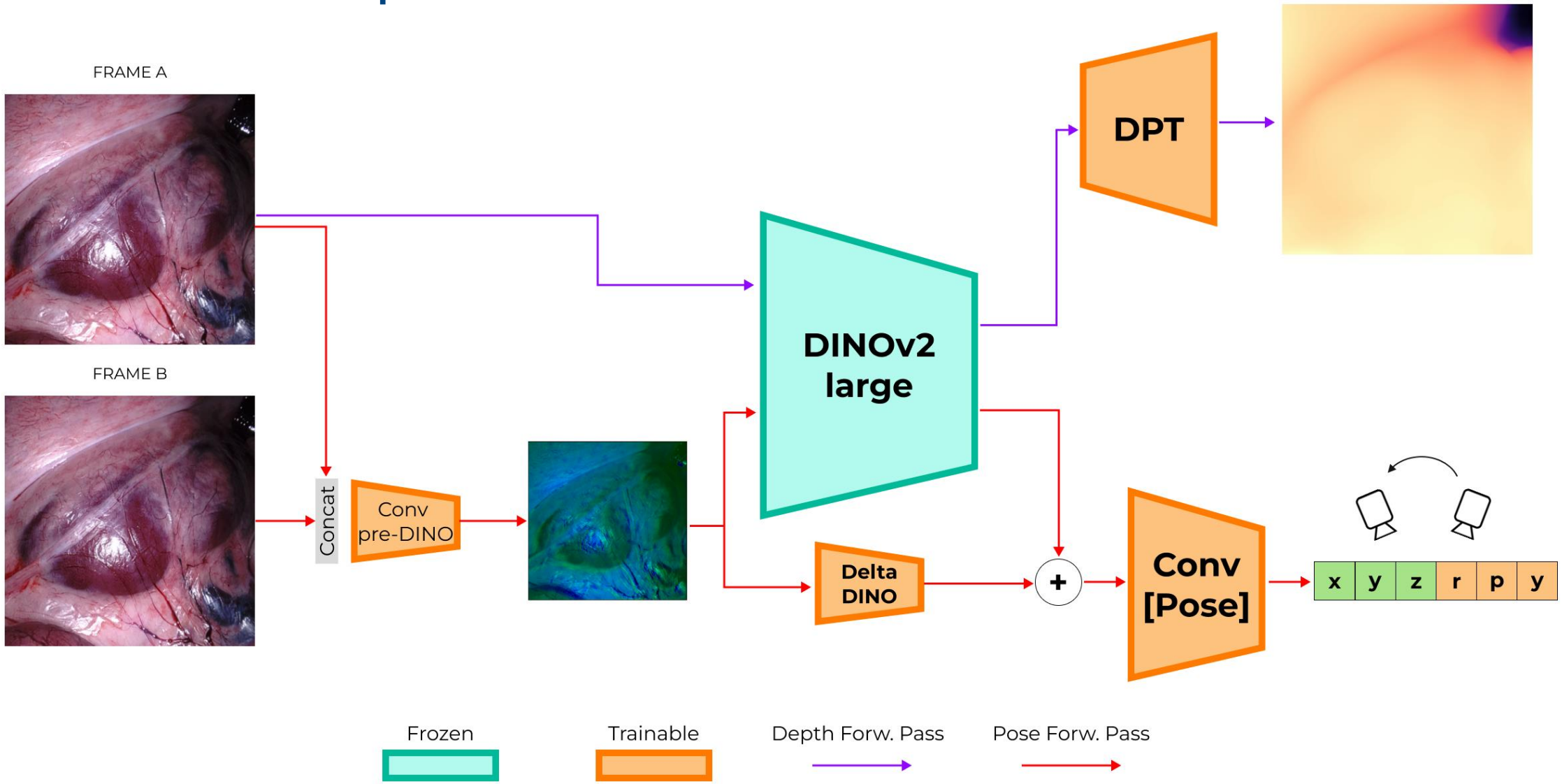
Architecture Updates



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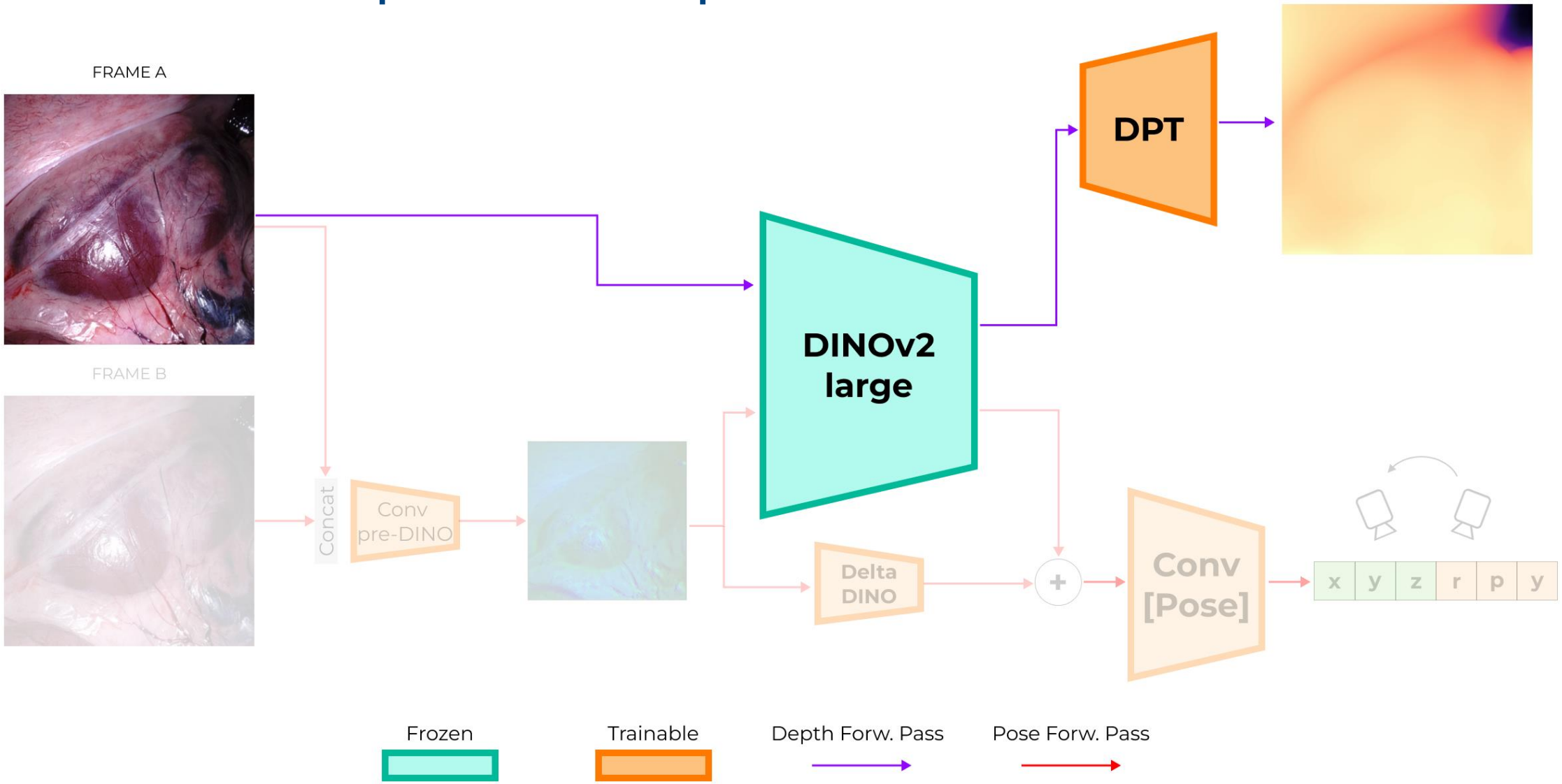
Architecture Updates [Depth]



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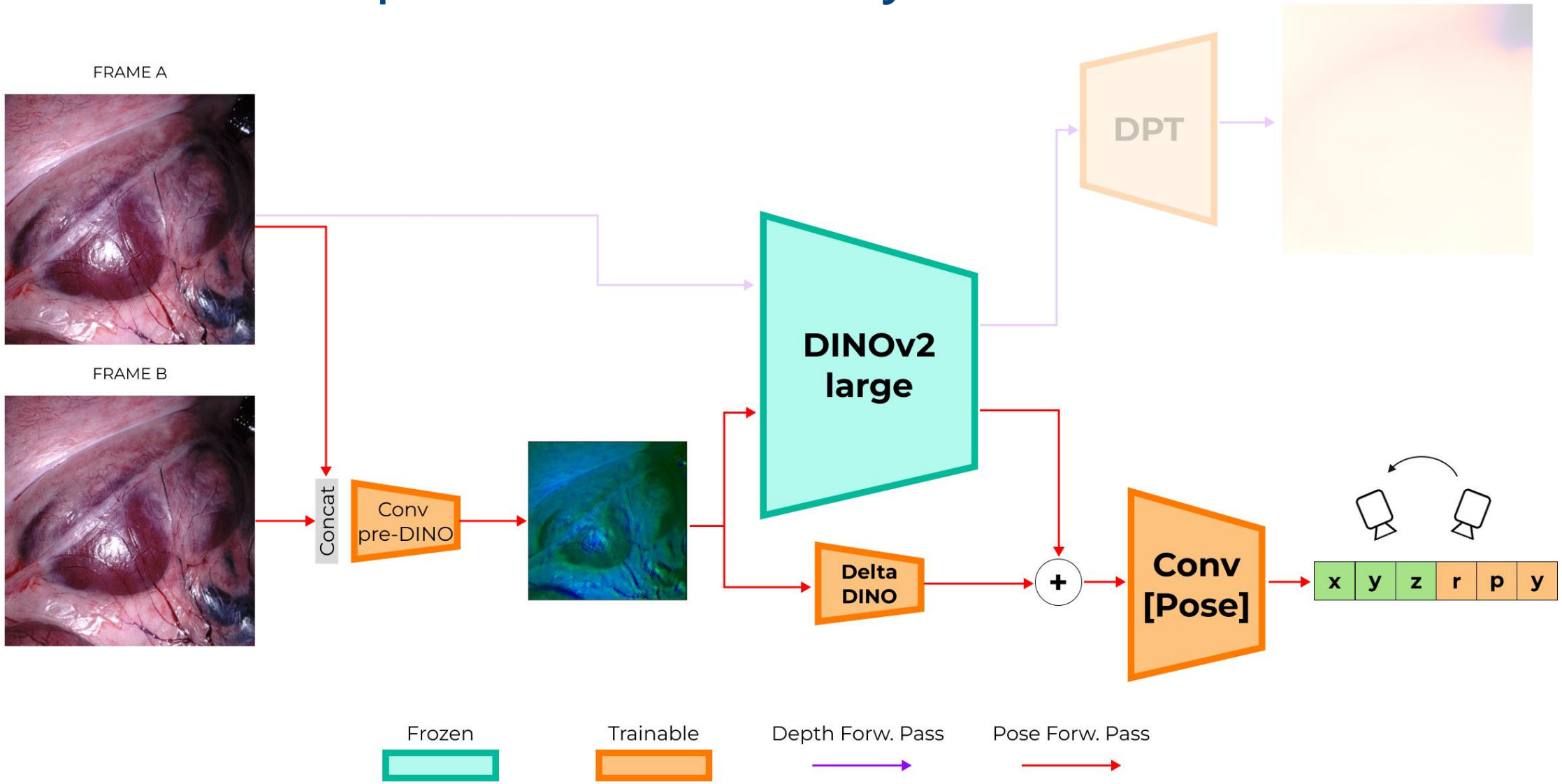
Architecture Updates [Odometry]



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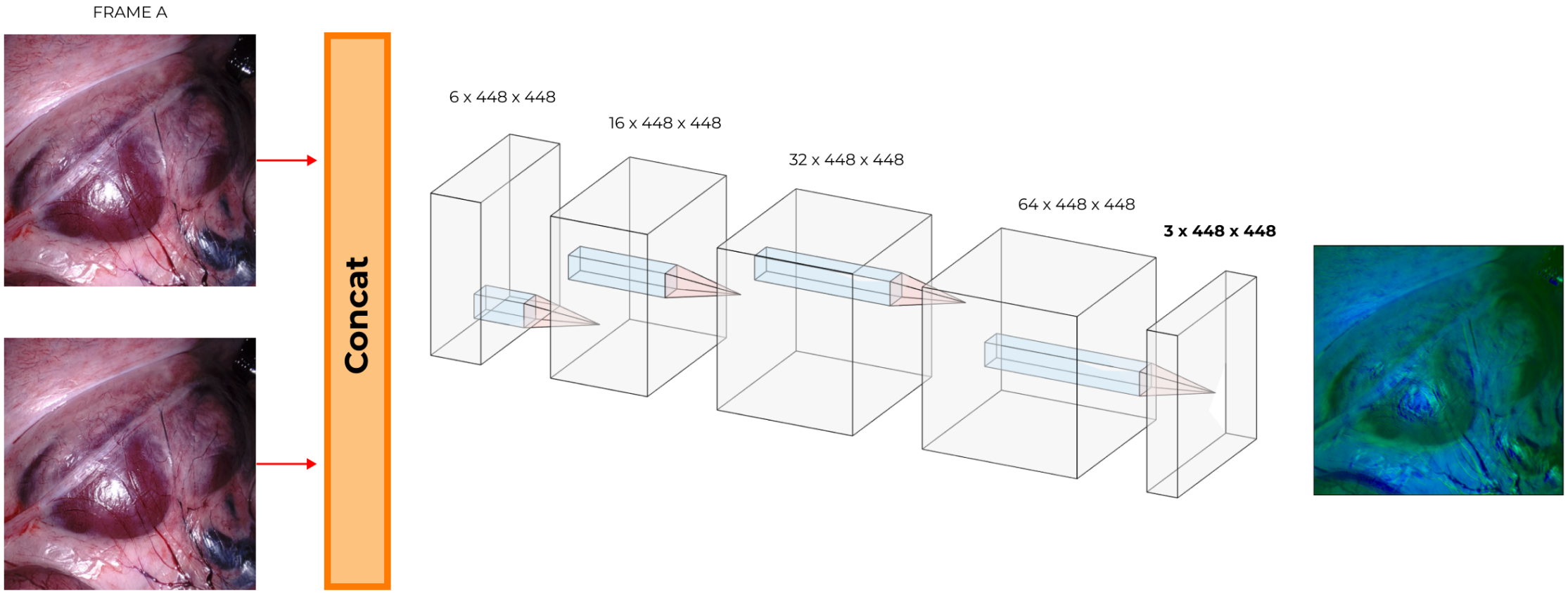
Architecture Updates [Pre-DINO]



Pre-DINO stage: Converting a pair of frames into a single RGB image interpretable by DINO and DeltaDINO

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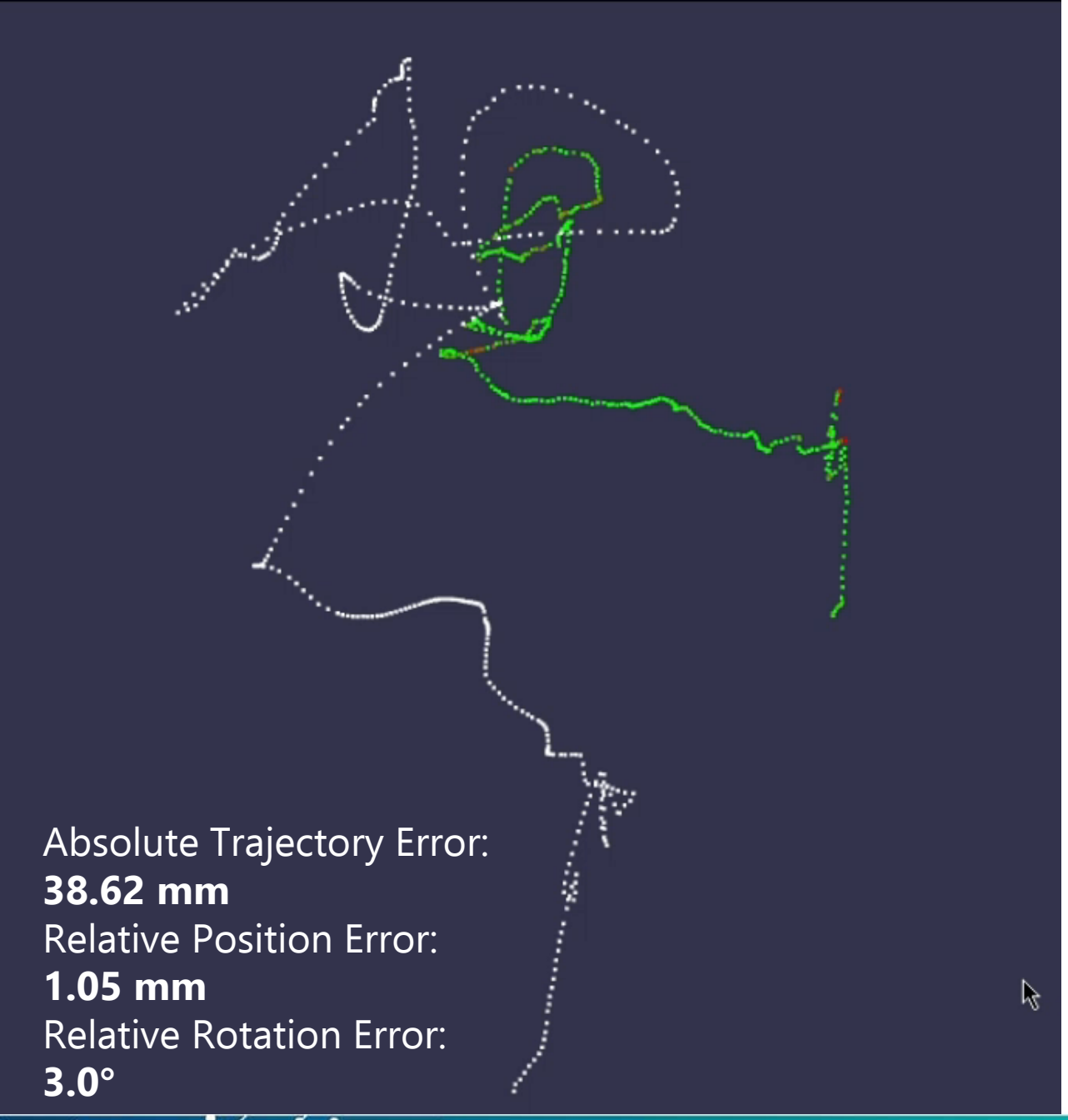

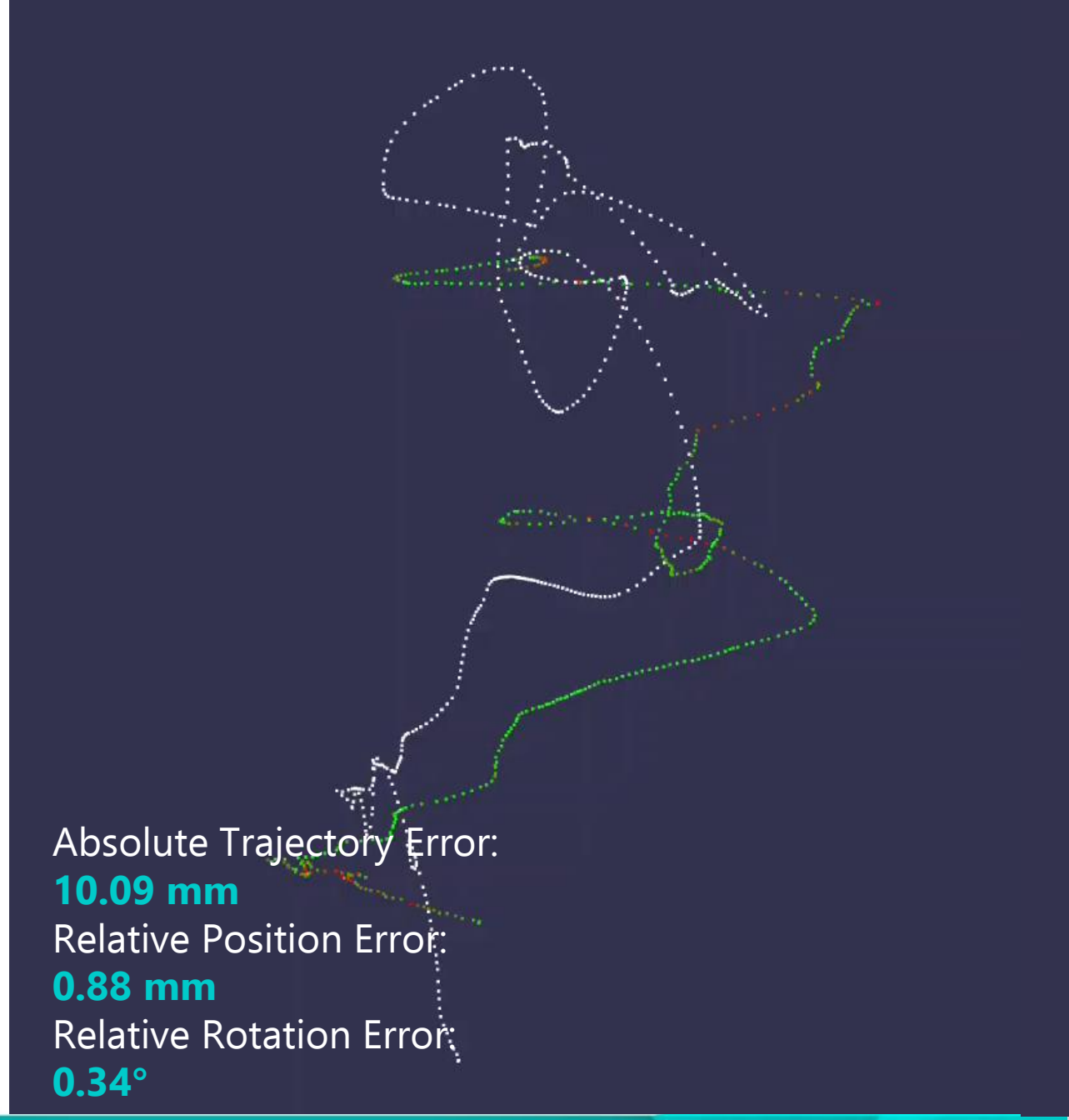

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WIP: Prediction Stabilization

Assuming a rigid scene, **all predictions (Depth & Odometry) should reproject to the same RGB-XYZ point cloud**

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Assuming a rigid scene, **all predictions (Depth & Odometry) should reproject to the same RGB-XYZ point cloud**

- At **training time:**
  Penalize differences between point clouds in the same batch

$$\mathcal{L}_{CHAM}(C_1, C_2) = \frac{1}{|C_1|} \sum_{p_1 \in C_1} \min_{p_2 \in C_2} \left(\alpha \|x_1 - x_2\|^2 + \beta \|c_1 - c_2\|^2 \right)$$

- At **inference time:**
  Correct the predicted camera pose with the transformation between the two point clouds (calculated through ICP)
  → **Bundle Adjustment**

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Project Status

Training Phase 1

- Depth Estimation: Fine-Tune DPT decoder → **100%**
- Odometry Estimation: Fully-Supervise train Decoder → **90%**

Training Phase 2

- Joint Depth + Odometry Estimation: → **50%**

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